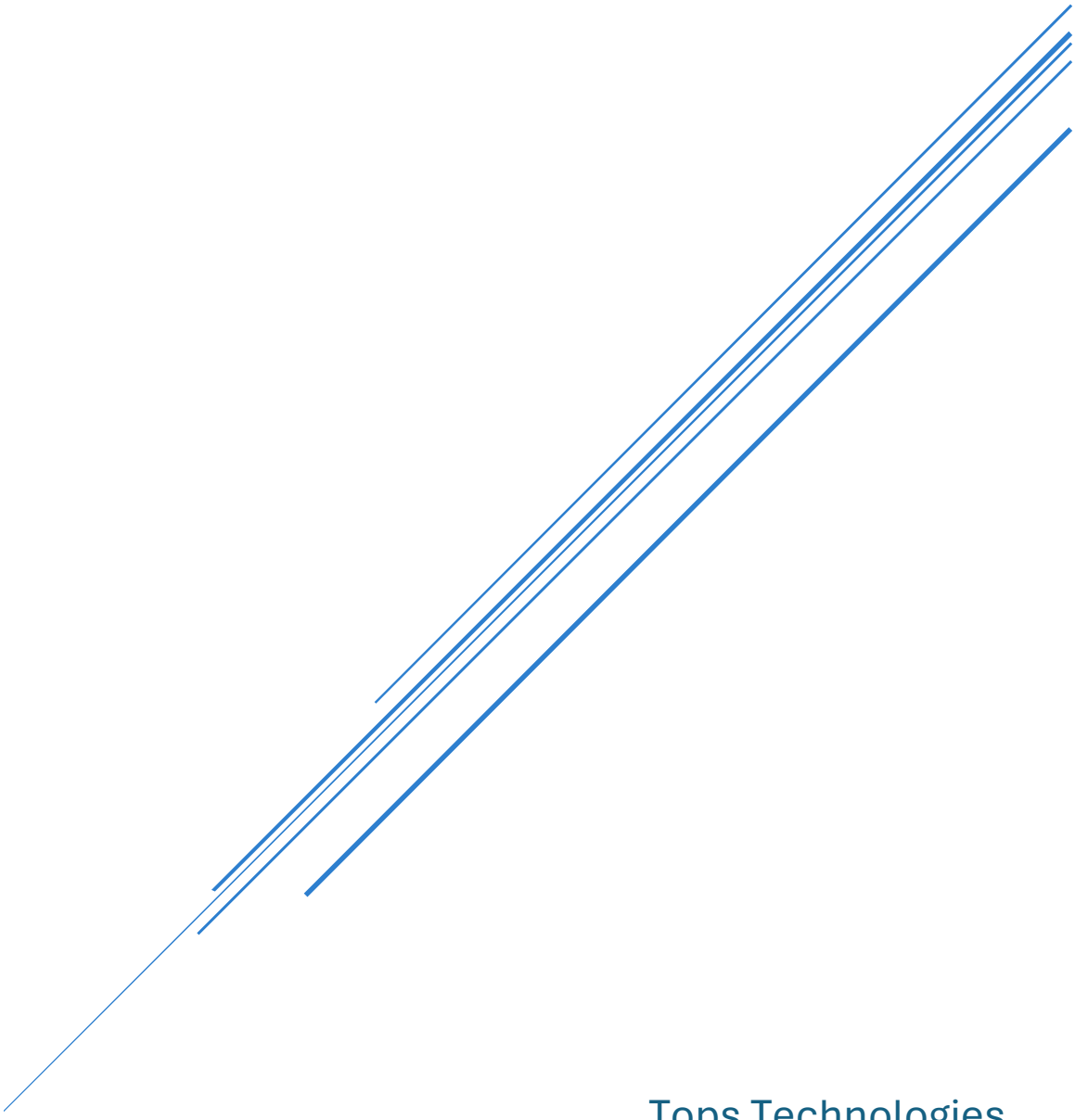
NETWORK TOPOLOGIES

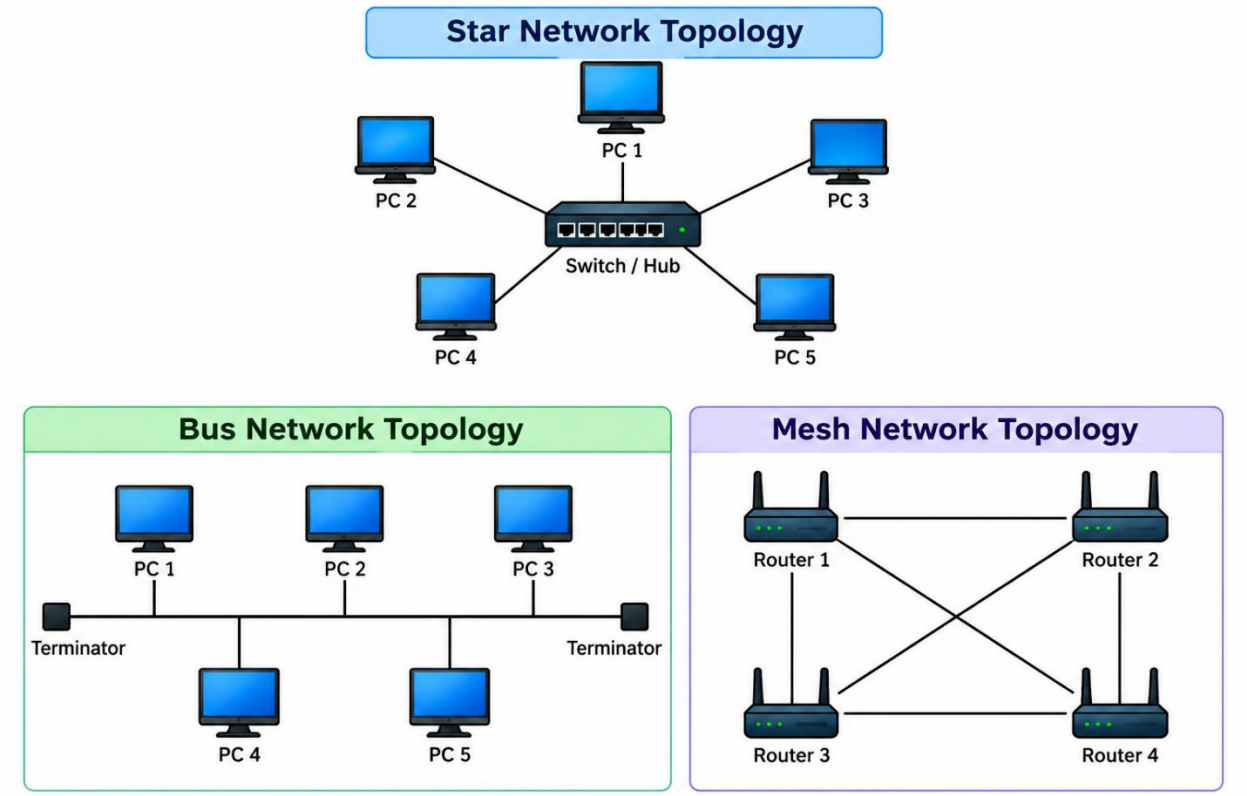
Assignment - 2



Tops Technologies

1. **Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.**

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I. Star Topology

Ex – WhatsApp

- WhatsApp can use a star-like network within an office or local network, where devices connect through a central switch or Wi-Fi access point to communicate and access services.

II. Bus Topology

Ex – Small Office or Computer Lab

- In a small office or computer lab, a bus topology can connect several computers through one main cable. It is simple and cost-effective for a small network.

III. **Mesh Topology**

Ex – IPL Live Streaming

- A mesh-like network can connect multiple servers and data centers through different paths. If one connection fails, data may travel through another path, improving reliability.

2. **Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).**

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➢ Comparison of Network Topologies

Topology	Advantages (4 Pros)	Disadvantages (4 Cons)	Best Use Case	Example
Star Topology	1. Easy to install and manage. 2. Easy to detect faults. 3. Failure of one computer does not affect others. 4. New devices can be added easily.	1. Requires more cables. 2. Requires a central switch or hub. 3. Failure of the central switch stops the network. 4. Installation cost can be high.	Offices, schools, and computer labs.	Instagram: Office computers can connect through a central switch to access the internet and Instagram.
Bus Topology	1. Uses less cable. 2. Low installation cost. 3. Simple to set up for small networks. 4. Does not require a central switch.	1. Main cable failure can stop the entire network. 2. Troubleshooting is difficult. 3. Performance decreases with heavy traffic. 4. Adding more devices can reduce performance.	Small or older computer networks.	Zomato: A small office lab could use this topology to connect computers for testing the application.
Ring Topology	1. Data travels in an organized direction.	1. A link failure can disrupt a basic ring. 2. Adding or removing devices can be difficult.	Some industrial and specialized networks.	Paytm: An office network could use a ring topology for internal communication; this does not mean Paytm itself uses it.

	2. Token-based rings can prevent data collisions. 3. Provides predictable data transmission. 4. Can work efficiently under steady network traffic.	3. Troubleshooting can be complex. 4. Data may pass through several devices before reaching its destination.		
Mesh Topology	1. Highly reliable. 2. Provides multiple paths for data. 3. Failure of one link may not stop communication. 4. Offers good fault tolerance.	1. Requires more cables and connections. 2. Expensive to install. 3. Difficult to configure and maintain. 4. Requires more ports and networking equipment.	Data centers, critical networks, and wireless mesh networks.	Instagram: Redundant network connections can help data centers maintain connectivity if one link fails.

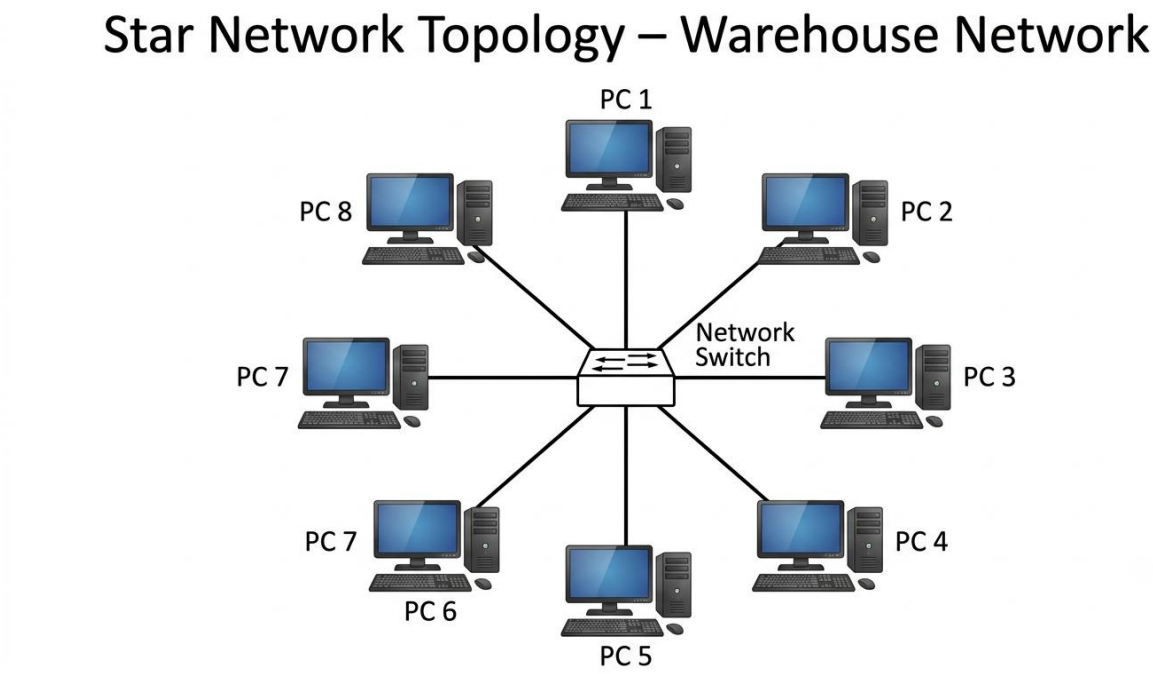
3. **Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.**

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I would choose a mesh topology for a multiplayer mobile game like PUBG or Free Fire because it provides multiple paths for data transmission between servers. It improves speed by allowing data to travel through suitable network paths and provides high reliability if one connection fails. However, it requires more networking equipment and connections, making it more expensive to install and maintain. This topology can help maintain stable gameplay and reduce interruptions during multiplayer matches.

4. **Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.**
   Hint: Consider factors like cost, ease of troubleshooting, and future expansion.

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Star topology is suitable because it is affordable, easy to manage, and makes troubleshooting simple. If one computer or its cable fails, the other computers continue working. New computers can also be added easily when the warehouse expands.